Electromagnetic Communication

Prof. Dr. Ali Kara

Objective

learning basic concepts in electromagnetics and communications to understand radar and electronic warfare (EW) systems

Learning Outcomes

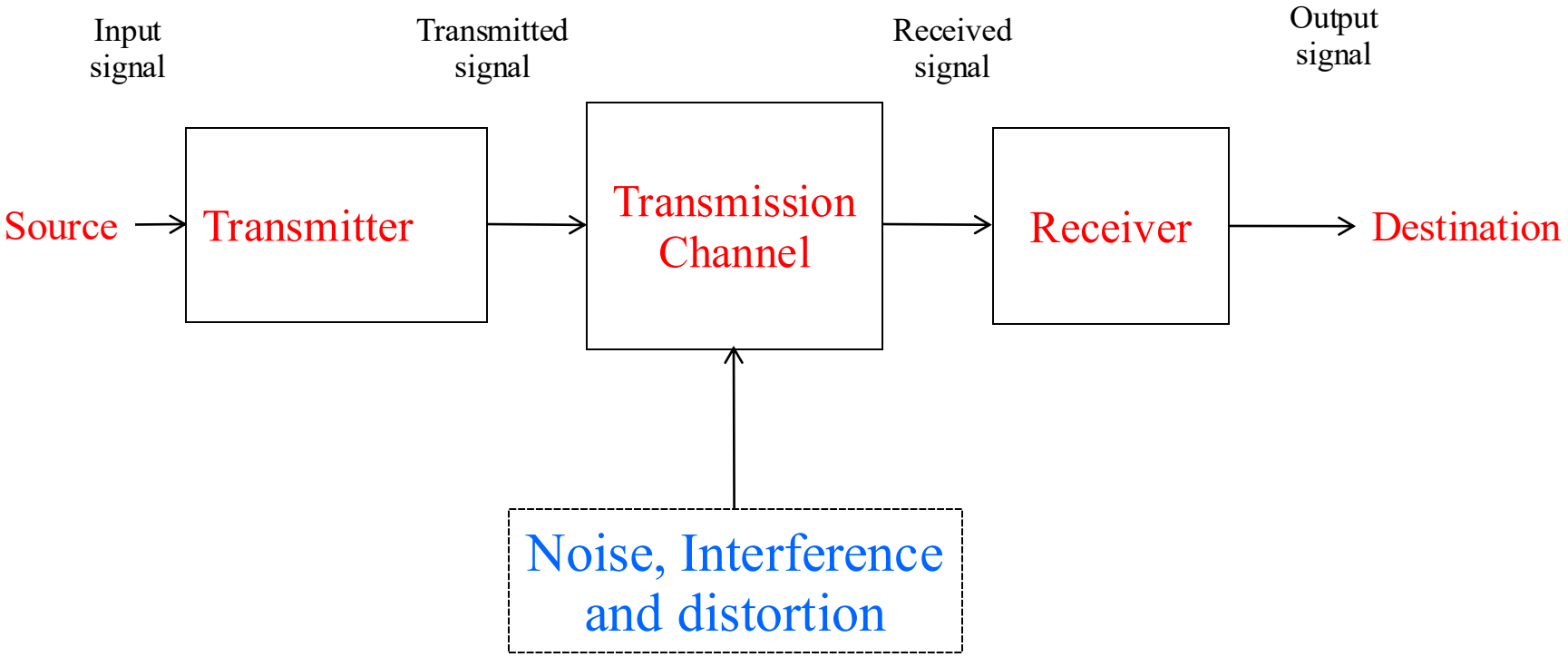
At the end of the lesson, you should be able to

1. describe a communication/radar signal
2. understand time and frequency domain analysis
3. describe an electromagnetic (EM) wave
4. describe basic parameters of EM waves, and propagation mechanisms
5. describe basic parameters of antennas
6. describe noise in communication/radar receivers

Presentation Plan

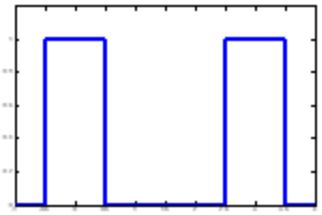
- Communication Systems
- Modulation (spectral and temporal)
- Radar type signals
- Electromagnetic waves, radiators, path loss
- Polarization, Radio propagation
- Antennas and main parameters
- Noise figure

Communication system

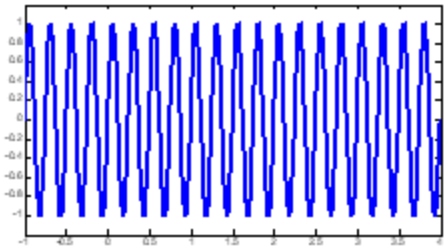


Modulation (temporal)

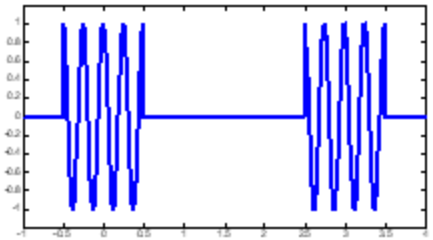
- Message signal= modulating signal; the signal to be transmitted
- Carrier signal = the signal to be used to transmit the message signal (auxiliary signal)
- A mplitude Modulated signal=(message signal) x (carrier signal)



Message

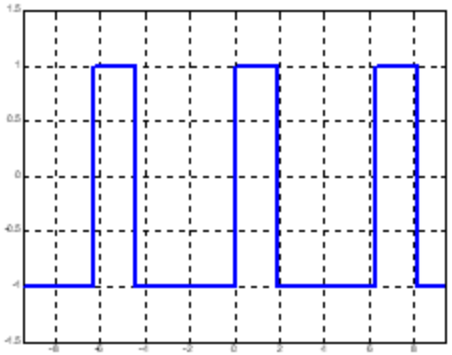


Carrier



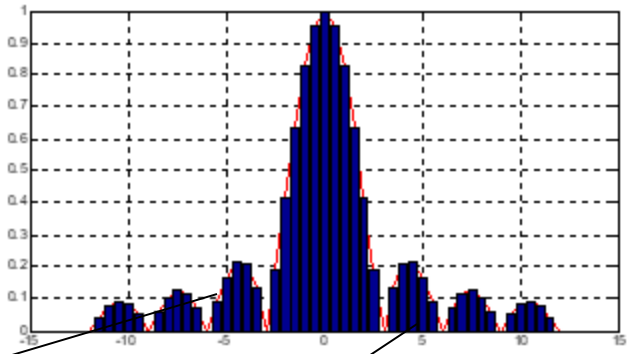
Modulated signal
(ASK or radar signal)

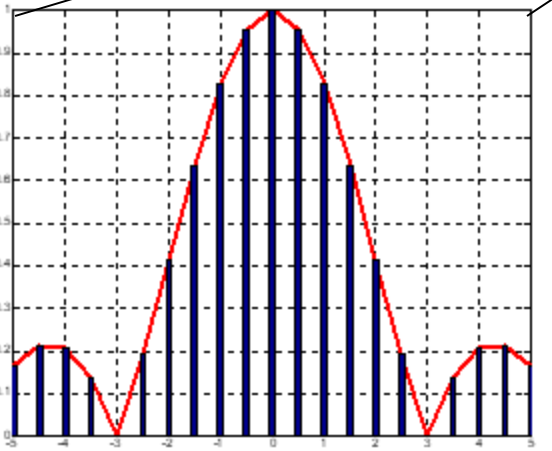
Pulse Train



temporal

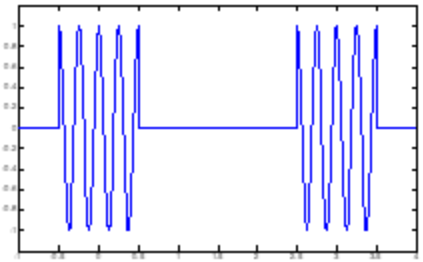
Fourier Analysis





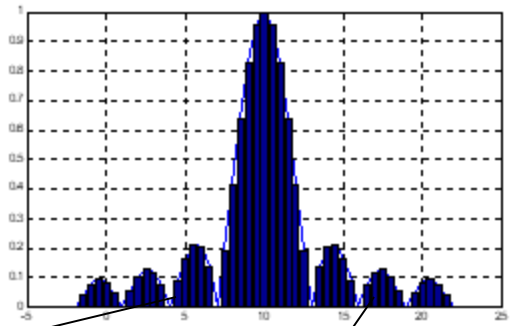
spectral

Modulated Pulse Train (ASK)

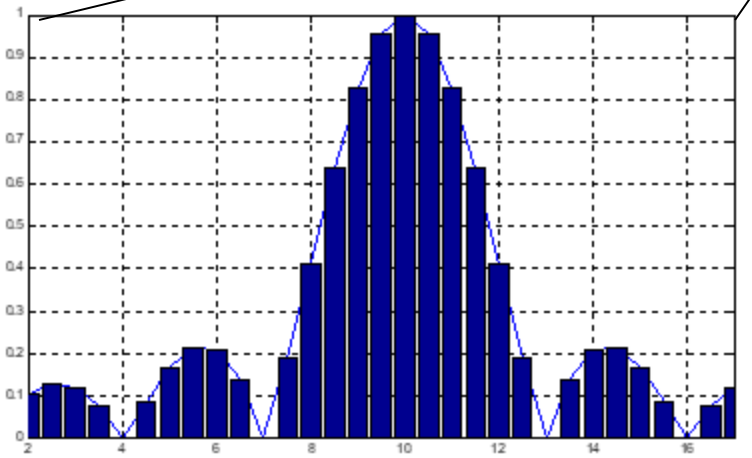


temporal

Fourier Analysis

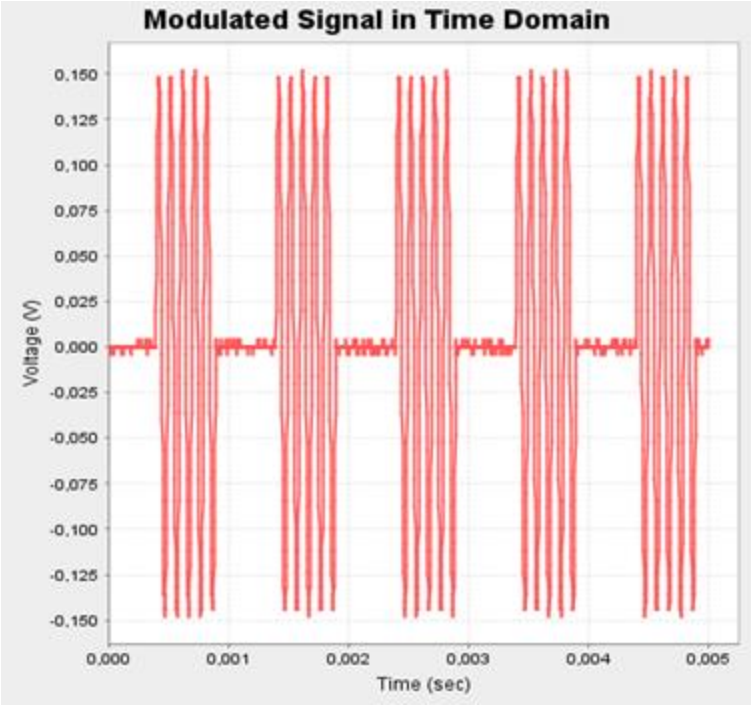


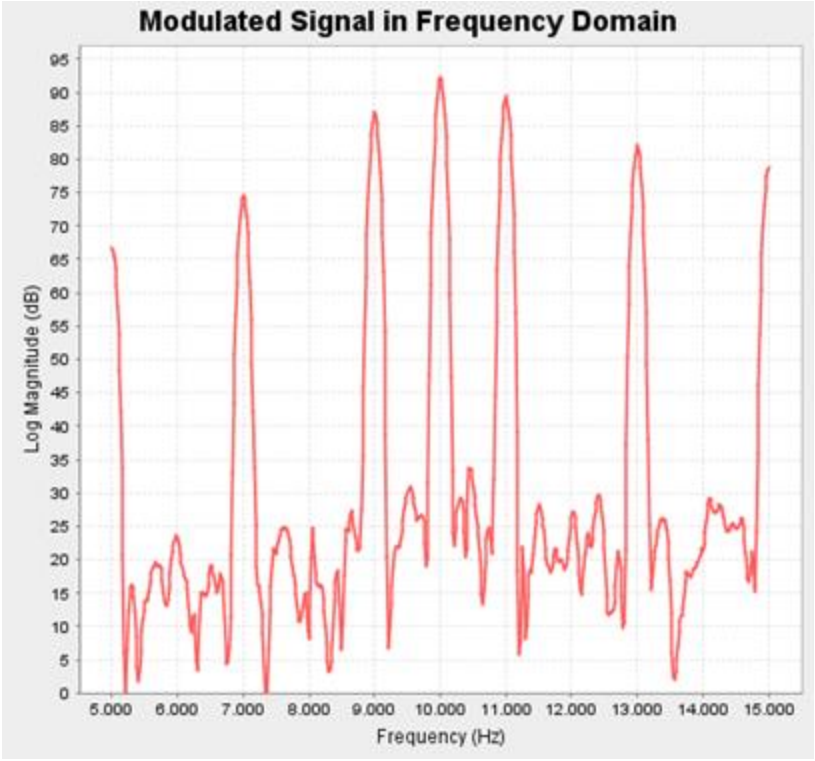
spectral



shifted spectrum

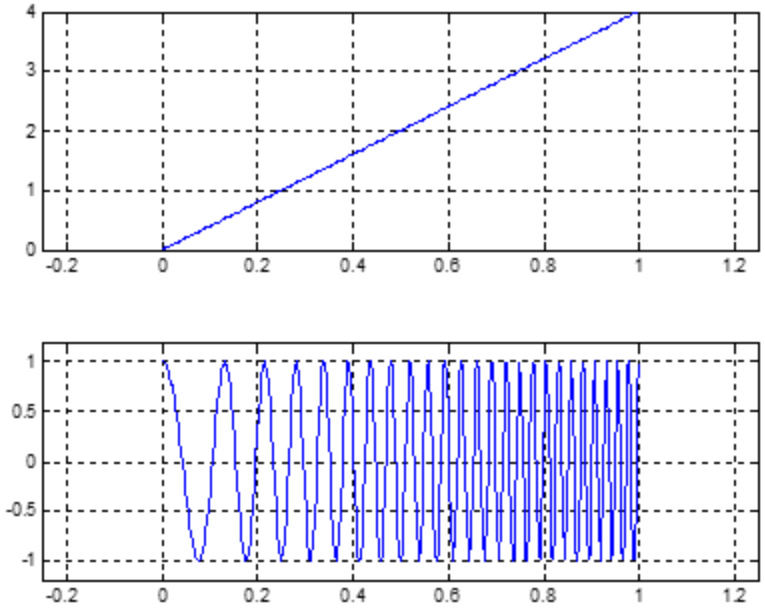
ASK: experiment (ERRL)



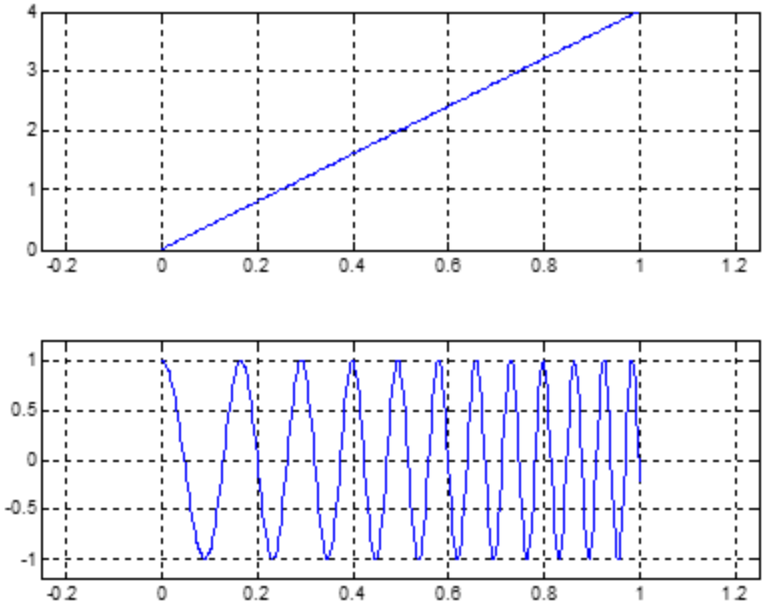


1kb/s
f=10kHz

Analog Modulations: FM &PM

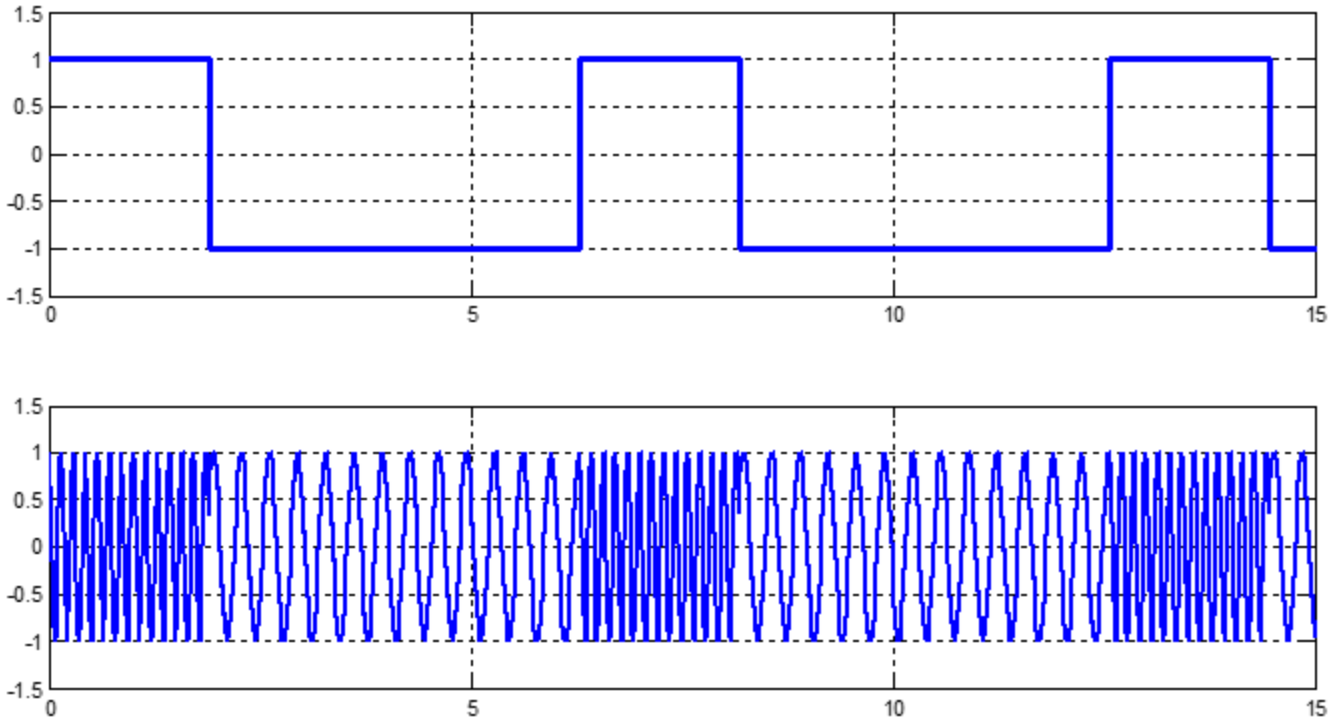


FM

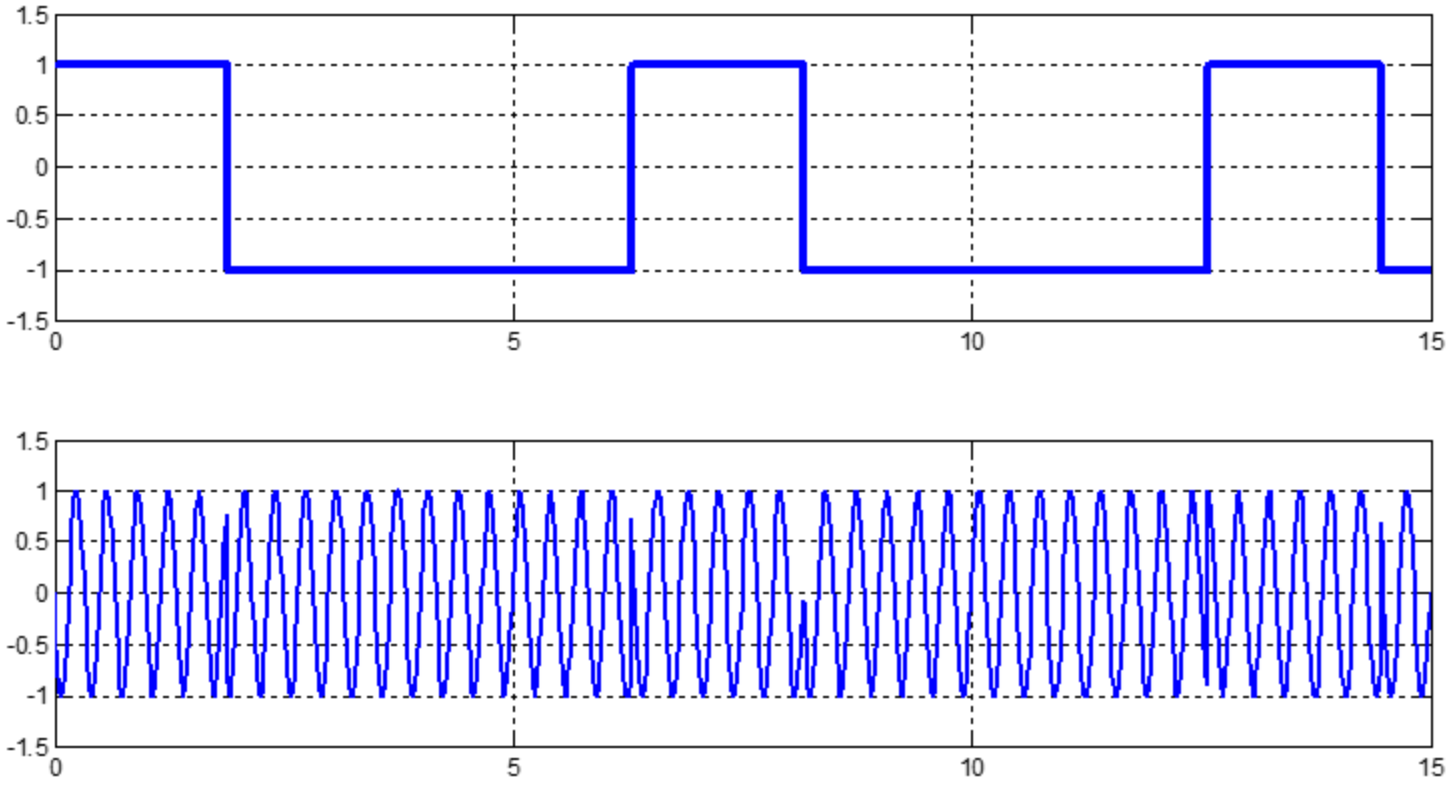


PM

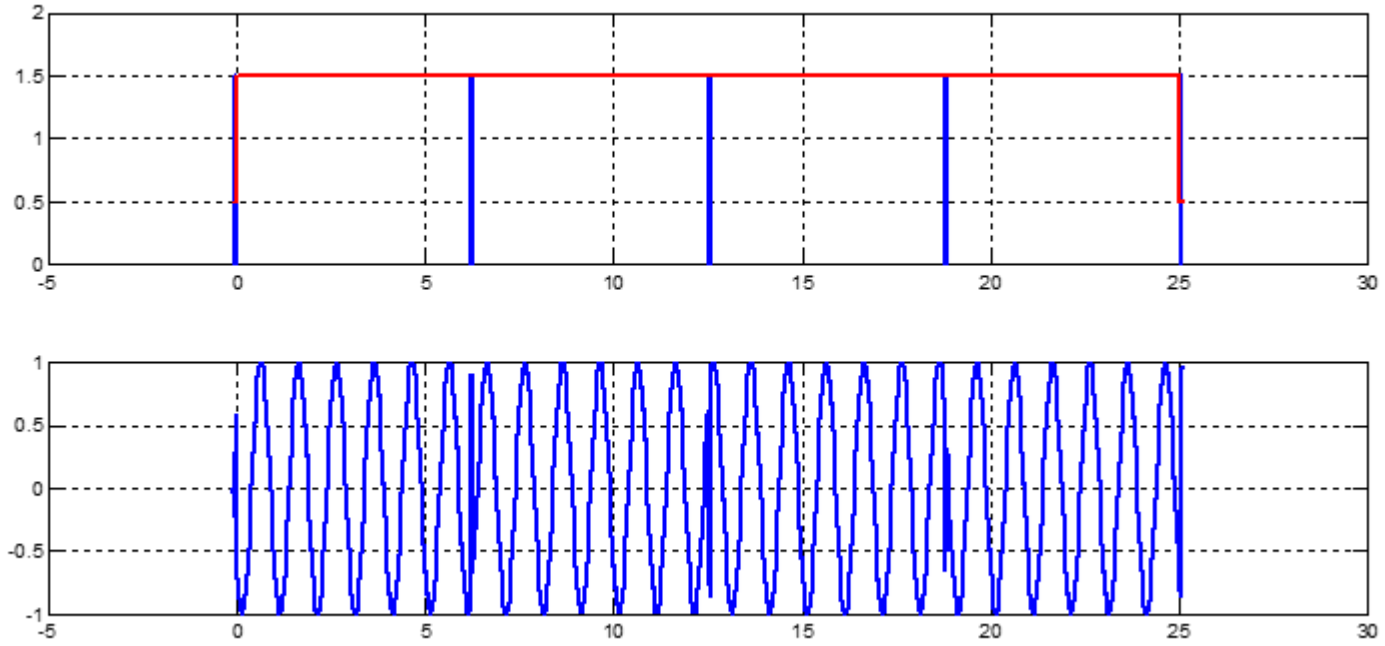
Digital Modulations: FSK



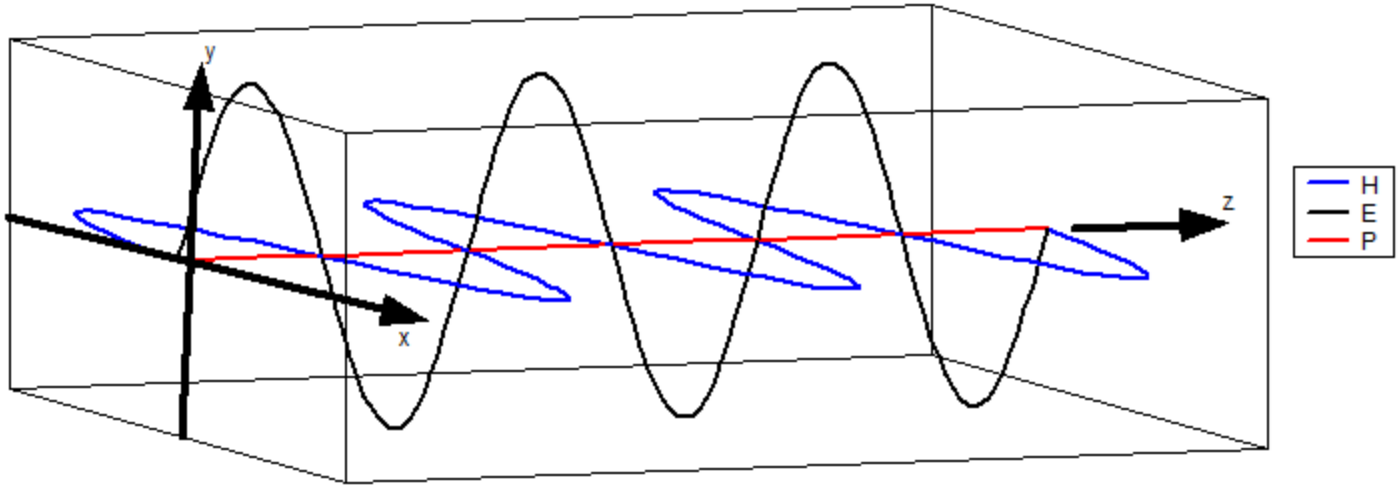
Digital Modulations: PSK



Modulation on Pulse (MOP): PMOP



EM Waves (E,H and P)



According to the diagram, what is the relationship between the electric field (E), magnetic field (H), and the direction of wave propagation (P)?

a) They are all parallel to each other.

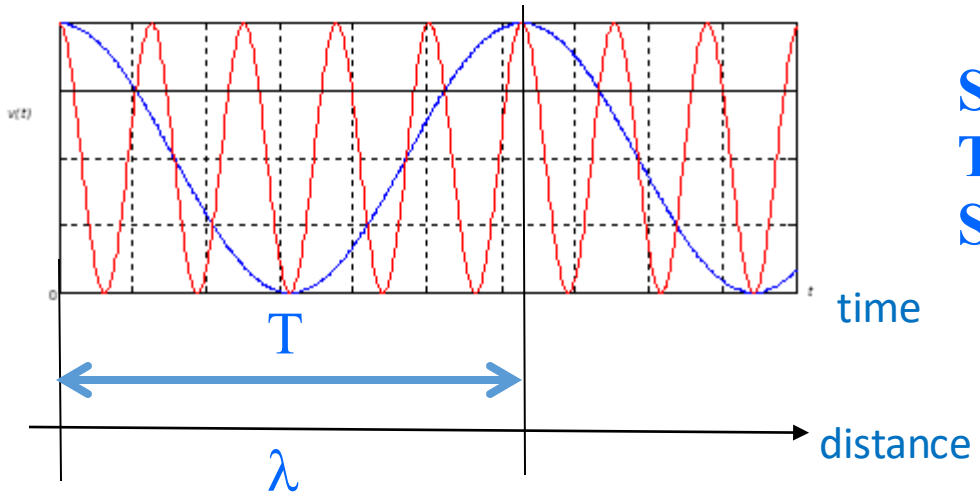
b) The electric field is parallel to the direction of propagation, and the magnetic field is perpendicular.

c) They are all mutually perpendicular to each other.

d) The magnetic field and electric field are in the same direction, perpendicular to the direction of propagation.

Frequency(f), Period (T) and Wavelength(λ)

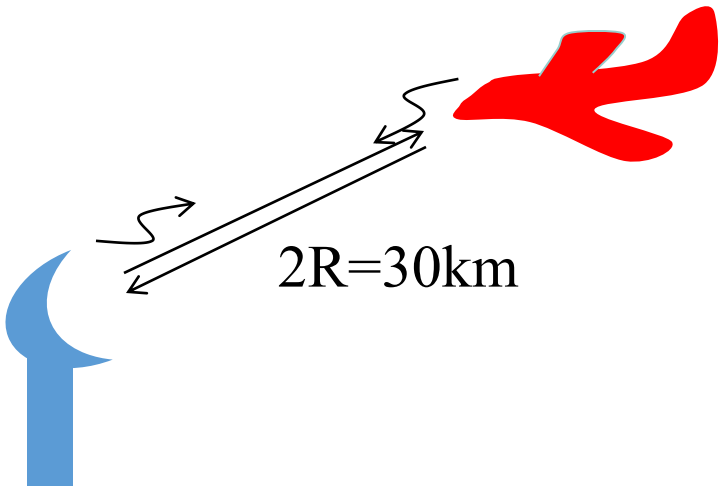
- Speed of EM waves=speed of light ≡ c=3 x 10^8m/s= f λ
- Frequency: f=c/ λ (spectral)
- Wavelength: λ=c/f (spatial)
- Period: T=1/f (temporal)



λ=c/f=cT

Spatial => wavelength
Temporal => period
Spectral => frequency

Example: speed of EM wave (Traveling time)



$$t=\frac{distance}{speed}=\frac{2R}{c}=\frac{30000m}{3\times10^8\,m/s}$$
$$=100\times10^{-6}\,s=100=0.1ms$$

$$v=2000\,km/\mathrm{h}=2000\frac{1000\mathrm{m}}{3600\mathrm{s}}=555.5\mathrm{m/s}$$
$$S=t\approx555.5\frac{\mathrm{m}}{\mathrm{s}}\;10^{-4}\mathrm{s}=0.0555m=5.55\mathrm{cm}$$

Physical and/or electrical length

- Size of a device, system or component in terms of the wavelength of operation
- Response of the device to EM waves of a specified frequency (wavelength)

Physical length of an aircraft L*=90m*

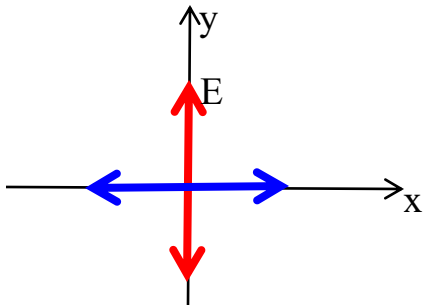
Operational frequency(RF) f*=3GHz*

Operational wavelegnth $\lambda = / f = 0.3/3 = 0.1m$

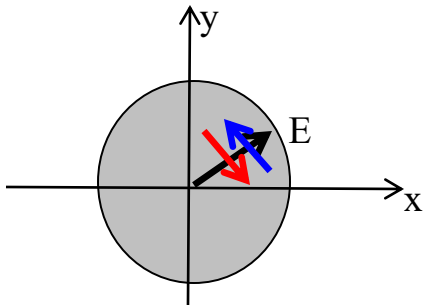
Electrical size of the aircraft $L = 90/0.1 = 900\lambda$

Polarization; types

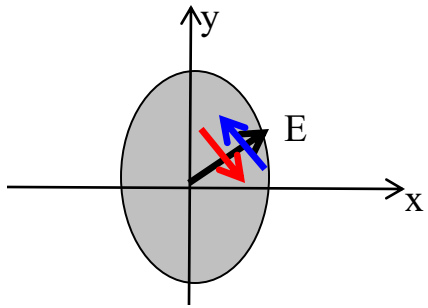
- EM waves are transmitted in two orthogonal dimensions, referred to polarizations
  - Horizontal and vertical polarization
  - Elliptical (RHC-right hand circular/elliptical or LHC-left hand circular/elliptical)
- Vertical polarization is dominant in most VHF/UHF applications
- When the direction of propagation is out of page (in z direction), then the tip of the electric field vector defines the polarization of the EM wave



Linear

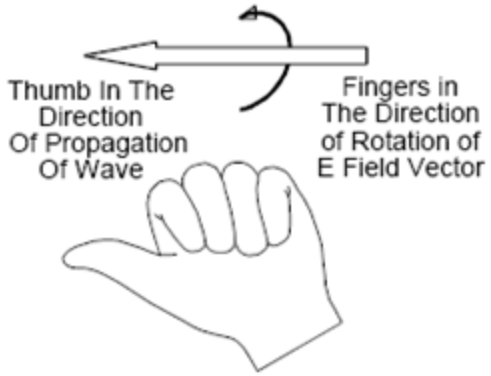


Circular

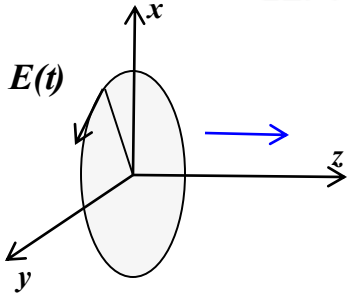


Elliptical

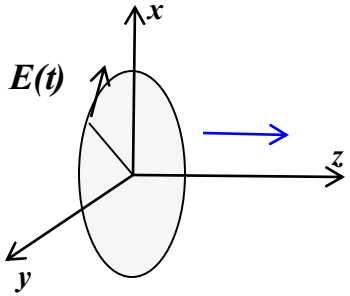
Polarization: Left-Right hand



LEFT HAND POLARIZATION

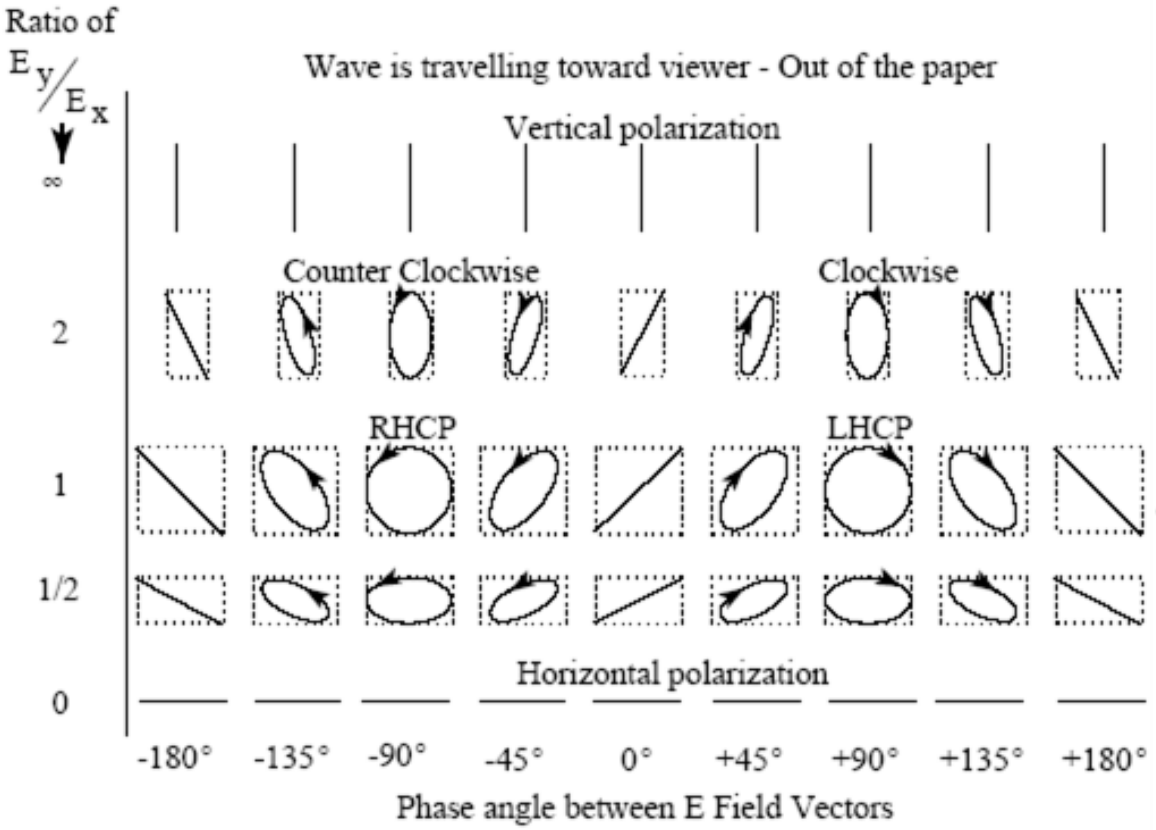


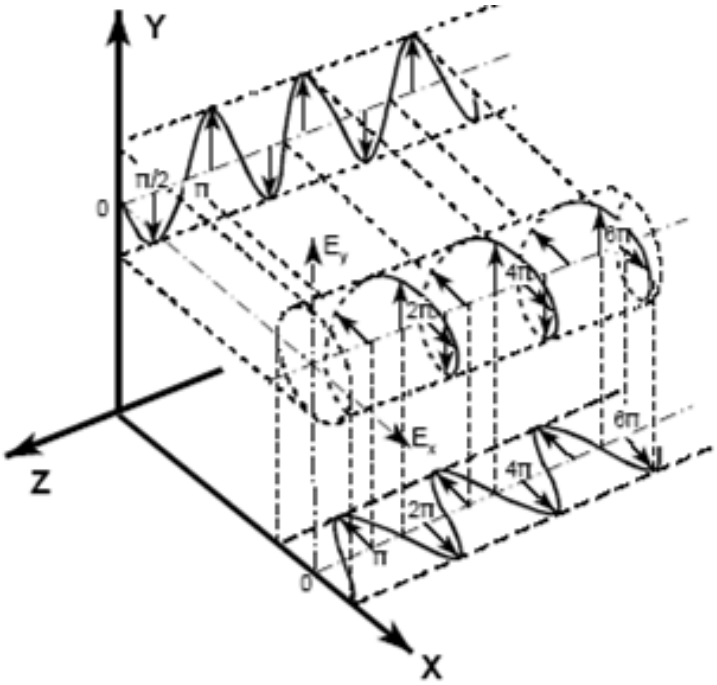
Right hand polarized



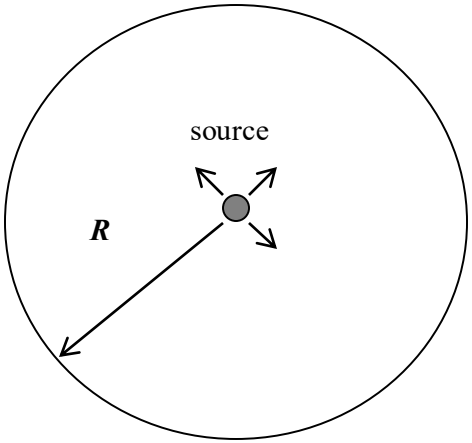
Left hand polarized

Polarization types





Reference source: Isotropic radiator



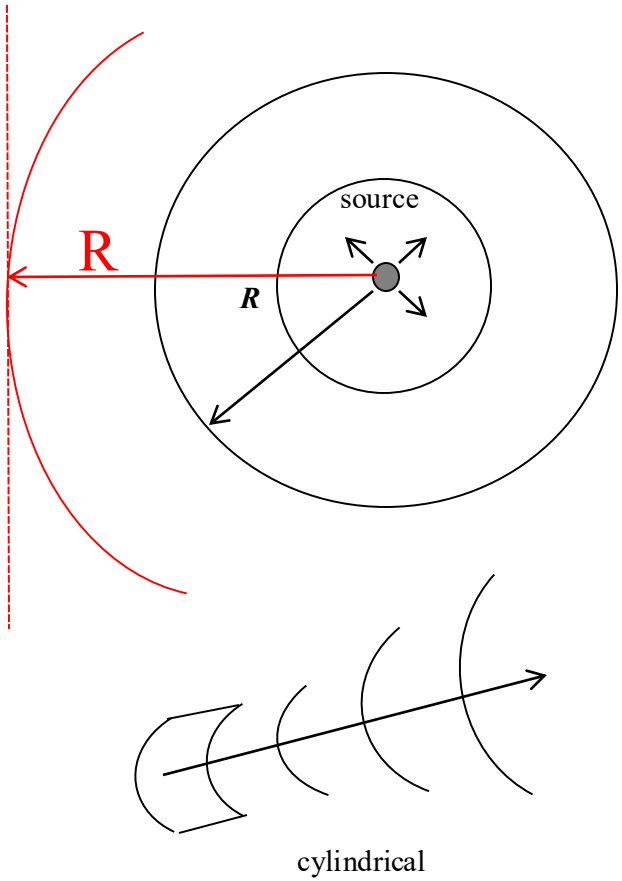
P_t :transmitted power,

P_d :power density @ R of a spherical surface,

$$P_d = \frac{P_t}{4\pi R^2}$$

Fictitious antenna !

Wavefront ⇔wave types



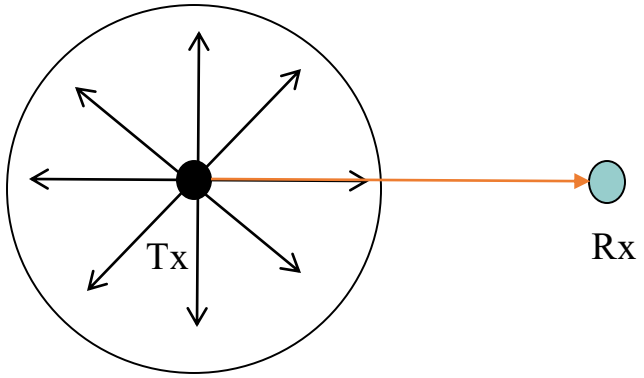
- Point source: radiating equally in all directions
- Phase is proportional to the distance from the source
- Wavefront: surface joining all points of equal phase
- Major wavefront: spherical, cylindrical and plane
- When R=> large, points source =>plane wavefront
- Use wavefront to classify EM waves type: plane, cylindrical, conical and spherical etc.
- EW Emitter wavefronts: Plane, cylindrical, spherical

Free space propagation

Power *flux density for a point source radiating* P_T

$$P_d = \frac{P_t}{4\pi R^2} (W/m^2)$$

$4\pi R^2$ *is the surface area of a sphere.*



Isotropic radiation

Relationship *between the Tx and Rx power for isotropic antennas*

$$P_R = \frac{P_T}{L_p} =_T \left(\frac{\lambda}{4\pi R}\right)^2 \text{ where path loss } L_p = \left(\frac{4\pi R}{\lambda}\right)^2$$

dB values

Linear number *X, Y and Z*

Power in watt, P and P', P"

Voltage in volts, V and V'

$$X(dB) = 10log_{10}(X)$$

$$X = 10^{X(dB)/10}$$

$$XY = Z$$

$$X(dB) + Y(dB) = Z(dB)$$

$$XP = P'$$

$$X(dB) + P(dB) = P'(dB)$$

$$X(dB) + P(dBm) = P'(dBm)$$

$$P/Y = P"$$

$$P(dB) - Y(dB) = P"(dB)$$

$$P(dBm) = 10log_{10}(P(w)/1mW)$$

$$P = 10W$$

$$P(dBm) = 10log_{10}\left(\frac{10W}{1mW}\right)$$

$$= 40dBm \ (= 10 \text{ dB})$$

$$P(dBm) - Y(dB) = P"(dBm)$$

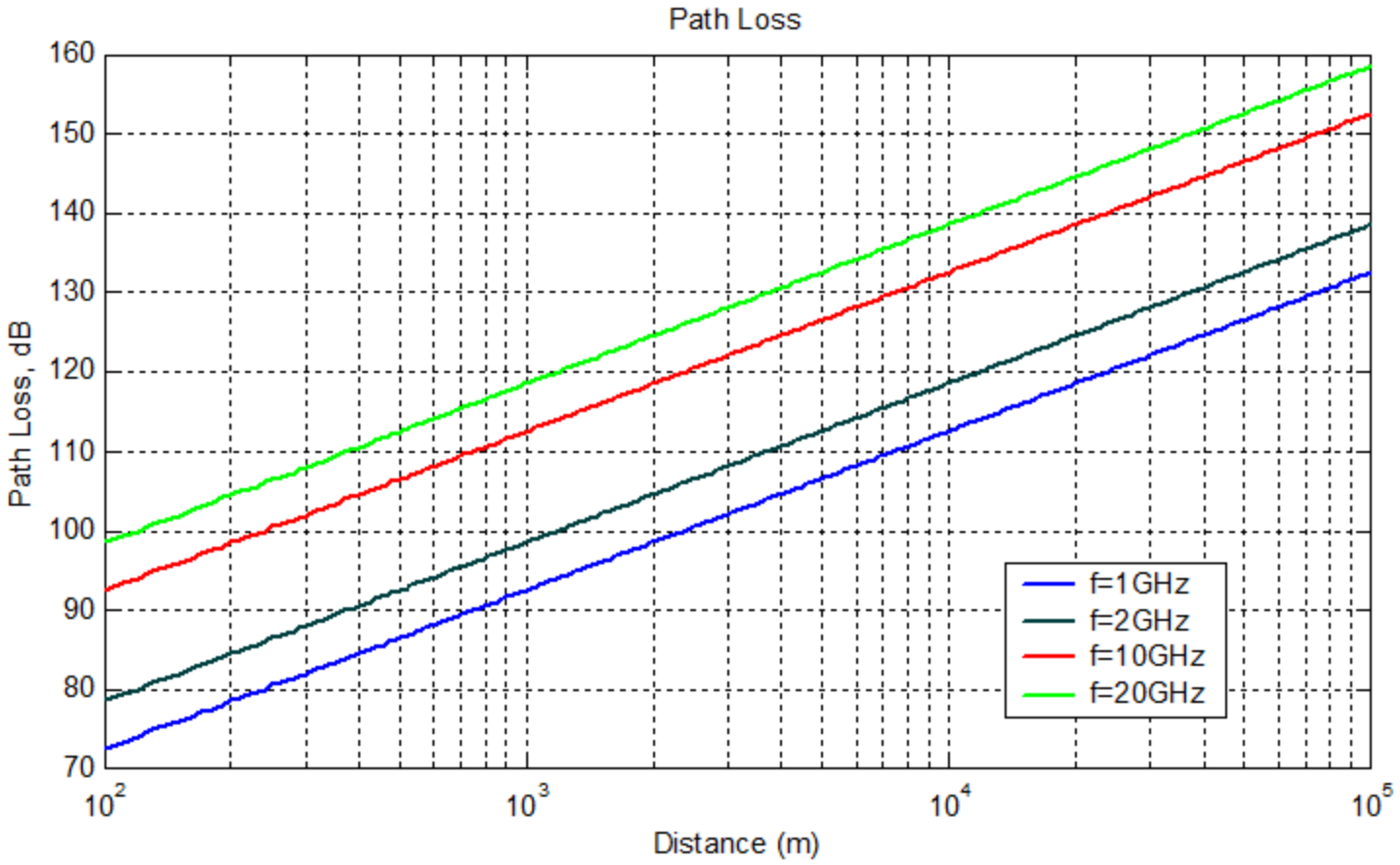
$$V(dB\mu V) + X(dB) = V'(dB\mu V)$$

$$\frac{P}{2}(dBm) = P(dBm) - 10\log(2) = P(dBm) - 3\ dB$$

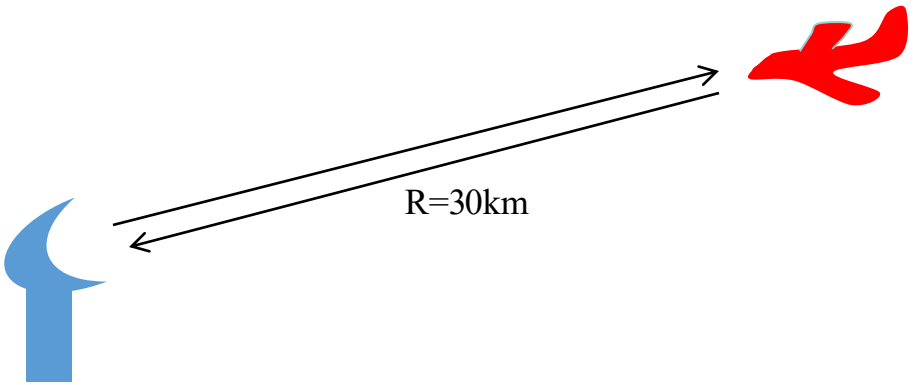
$$\frac{V}{4}(dBmV) = V(dBmV) - 20\log(4) = V(dBmV) - 12\ dB$$

Path loss and frequency

$$L_p = \left(\frac{4\pi R}{\lambda}\right)^2 \rightarrow L_p\,(dB) = 20\log\left(\frac{4\pi R f}{c}\right)$$



Example: Path Loss for a radar system



Surveillance radar:=1$GHz \Rightarrow \lambda = \frac{c}{f} = \frac{3\ 10^8\ m/s}{1 10^9\ Hz} = 0.3m$

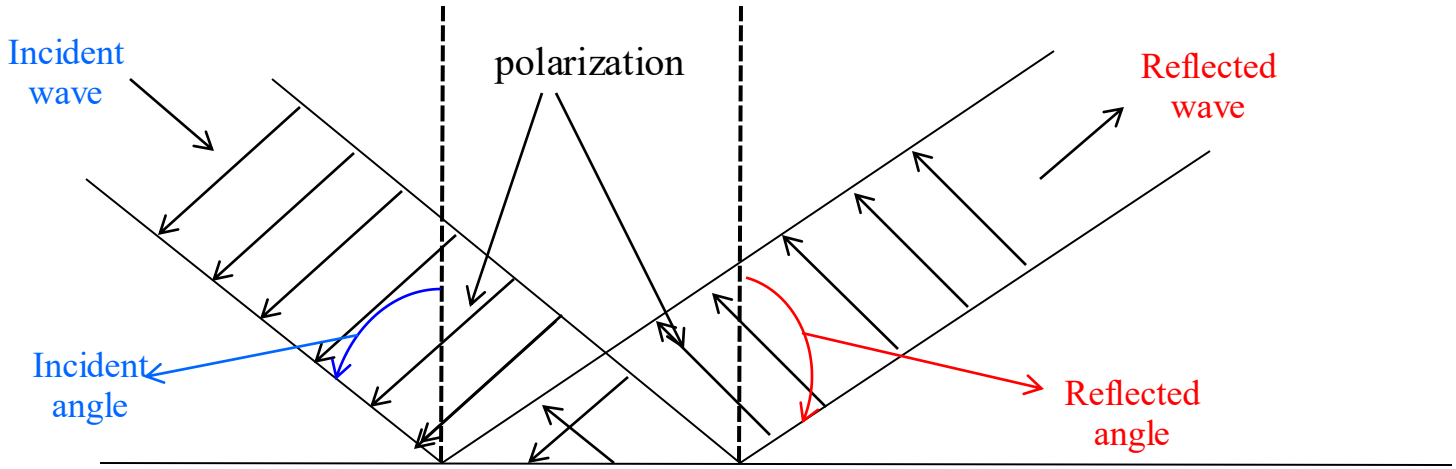
$L_p = 20log\left(\frac{4\pi R}{\lambda}\right) \approx 123B = 10^{123/10} = 10^{12.3}$ (times less power)

$P_r @argt = 10W \Rightarrow P_r = \frac{P_t}{L_p} = 10000 \times 10^{-12.3}\ W = 10^{-8.3}\ W = 5 \times 10^{-9}\ W = 5nW$

No amplification at the target, isotropic antennas !!!

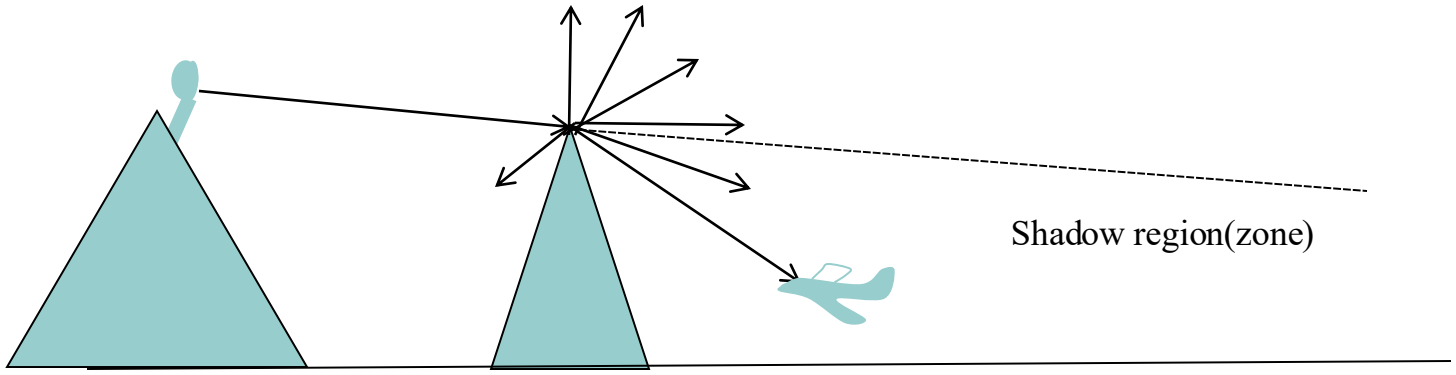
$P_r @adar = 5nW \times 10^{-12.3} = 5 \times 10^{-21.3} W$???

Reflection


Incident
wave
polarization
Reflected
wave
Incident
angle
Reflected
angle

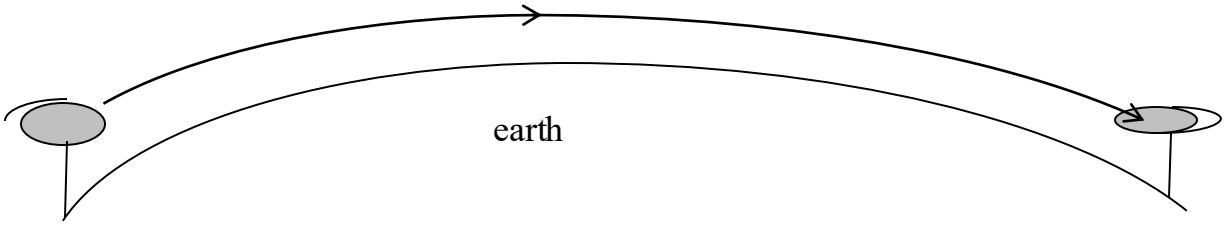
Diffraction

- when EM waves hit objects like hills and buildings,
- one portion of the wave are blocked (absorbed or reflected)
- remaining portion tends to bend and illuminate the shadow region



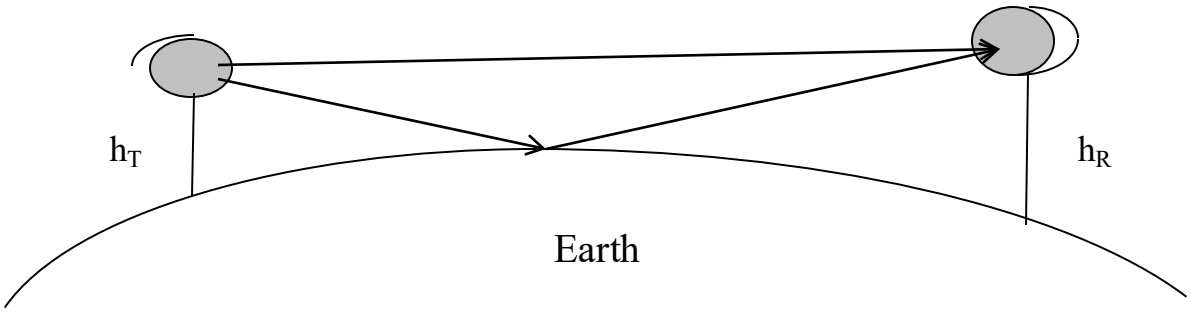
Ground wave

- Vertically polarized
- Changes of the terrain is important
- Attenuation ⇔surface impedance of the earth
- Much better over water (sea) than a dry surface

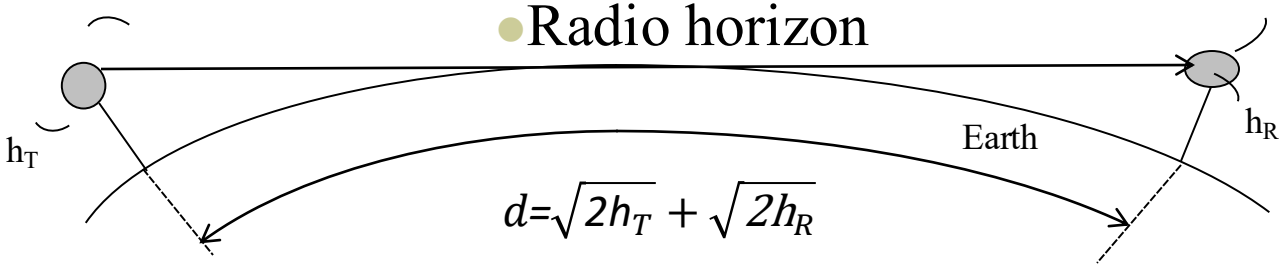


Space waves

- Direct wave: line of sight-LOS transmission (radars and satellites)
- Ground reflected wave

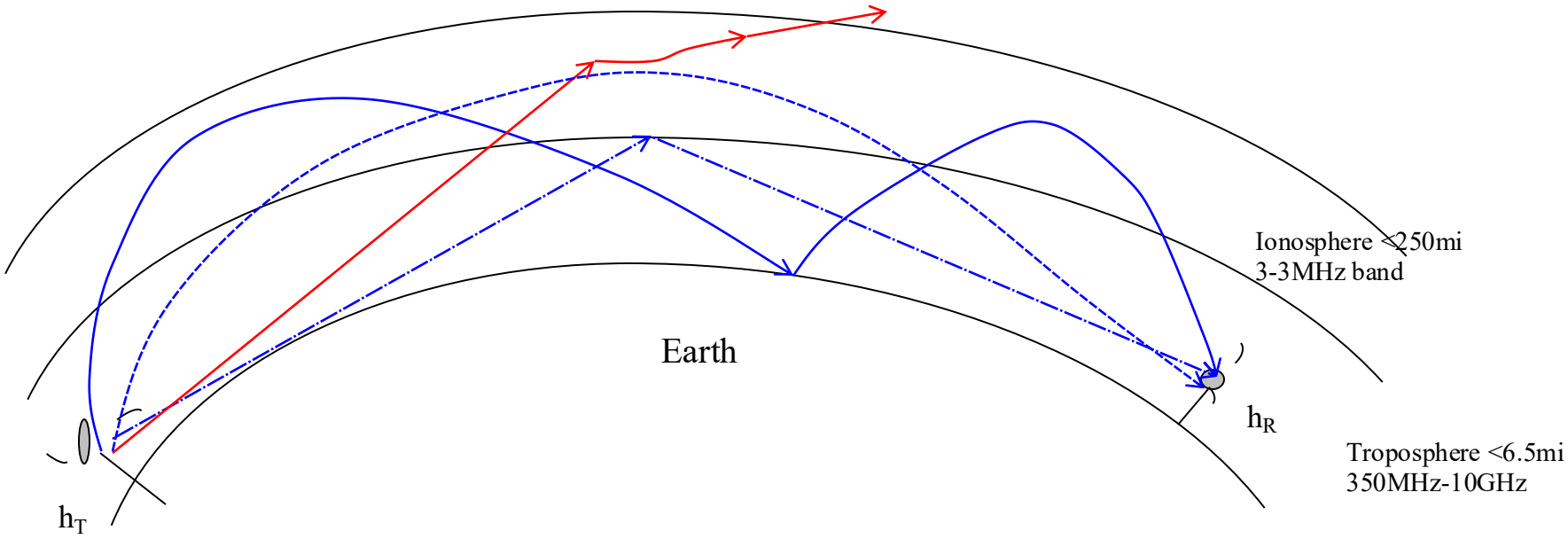


- Radio horizon



Sky Wave (The ionosphere)

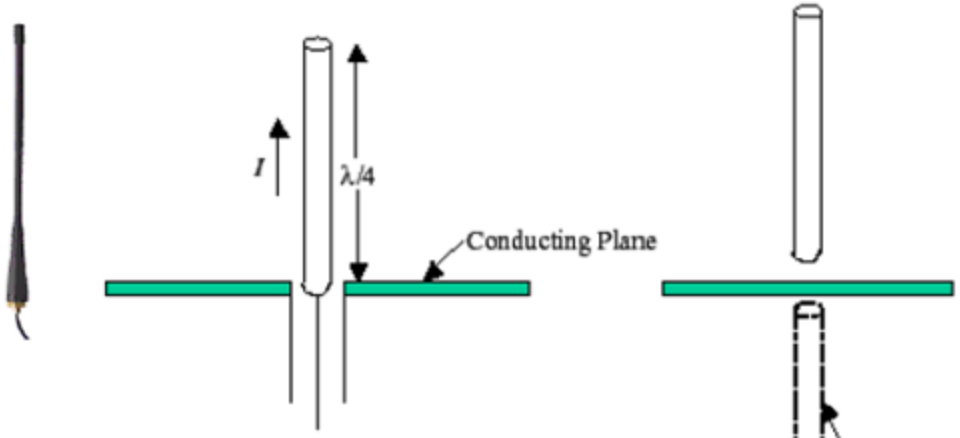
- Critical frequency
- Maximum usable frequency (MUF)
- Ionosphere layers (D, E, F_1,F_2)
- Day/night difference, seasonal differences



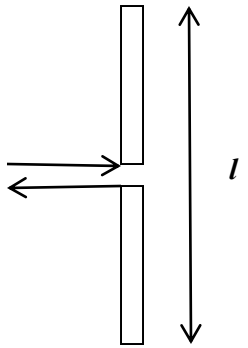
Antennas: Types

- Wire antennas (monopoles, dipoles, loops)
- Aperture antennas (sectoral horn, pyramidal horn, slots)
- Reflector antennas (parabolic dish, corner reflector)
- Lens antennas
- Microstrip antennas
- Antenna arrays

Wire antennas



Monopole



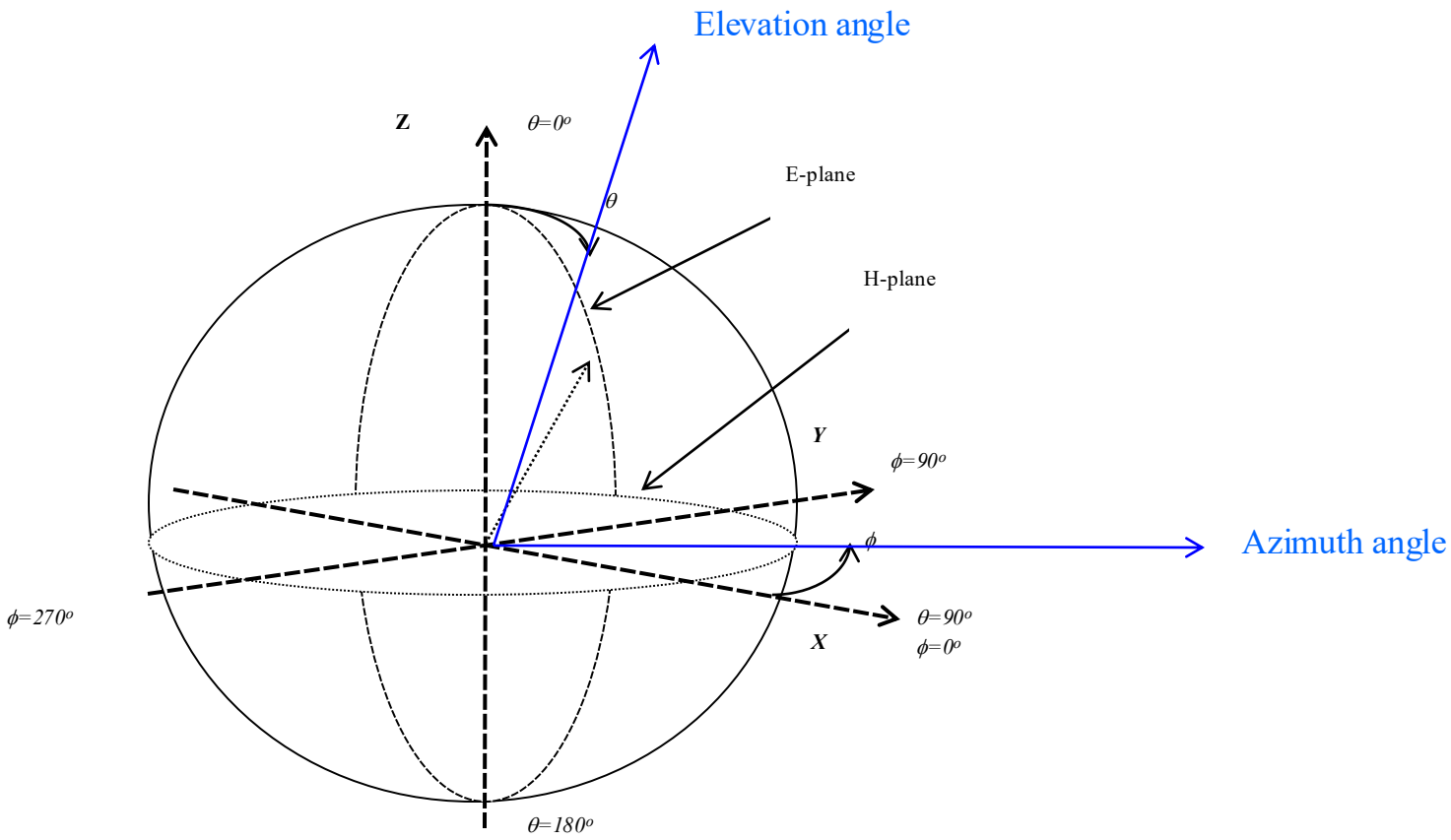
dipole

- Monopole, dipole, spiral, helix

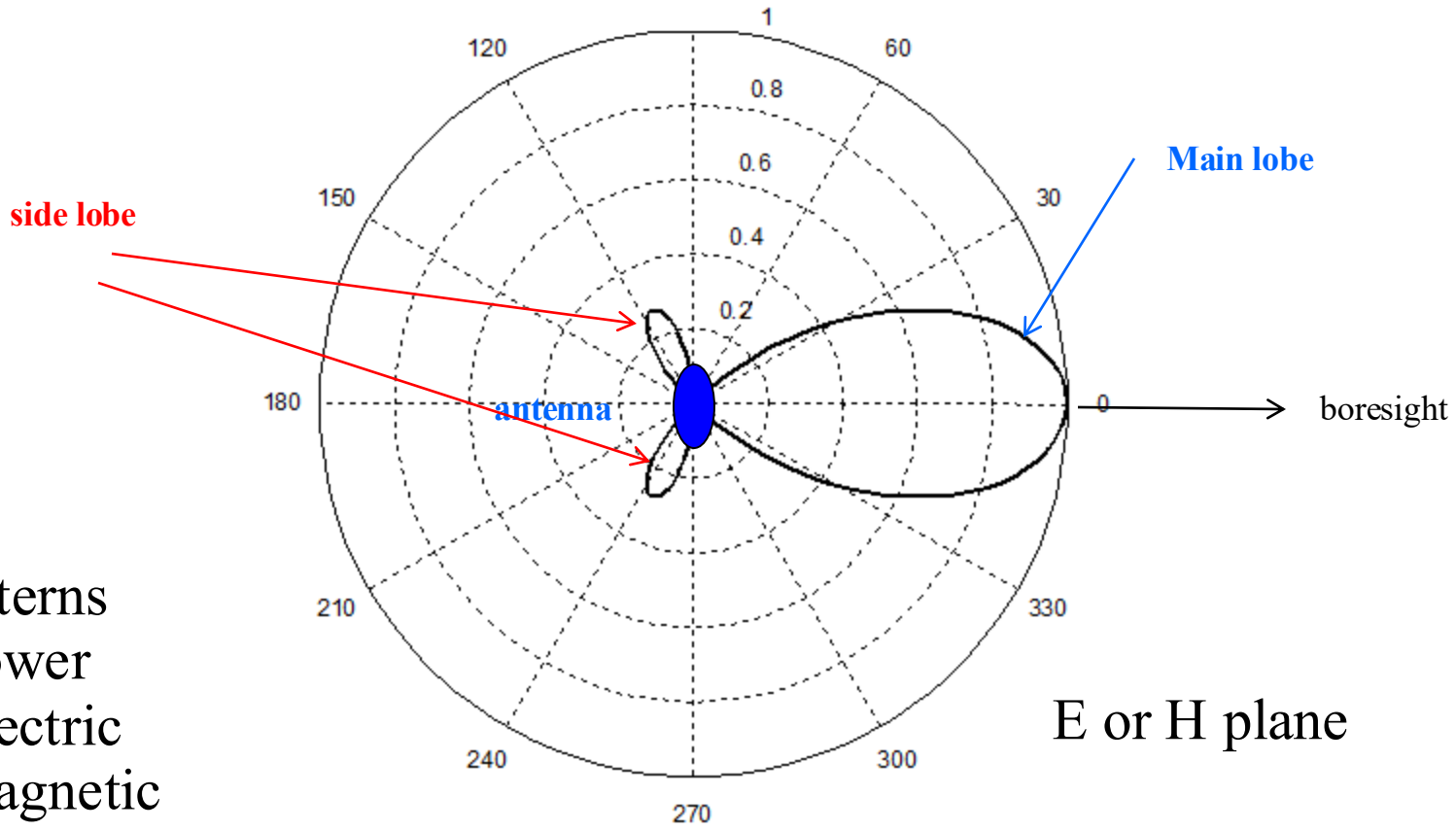
Antenna Parameters

- Radiation pattern - angular plot of the radiation intensity
- Omnidirectional pattern - uniform radiation in one plane
- Directive patterns - narrow beam(s) of high radiation
- Gain - directivity reduced by losses (≡Directivity*efficiency).
- Polarization - trace of the radiated electric field vector (linear, circular, elliptical).
- Bandwidth - range of frequencies over which performance is acceptable (resonant antennas, broadband antennas).
- Impedance - antenna input impedance at its terminals.
- Beam scanning - movement in the direction of maximum radiation by mechanical or electrical means.
- Other system design constraints - size, weight, power handling, cooling, cost, radar cross section, etc. (SWaP-C)

Spatial domain: positioning in space



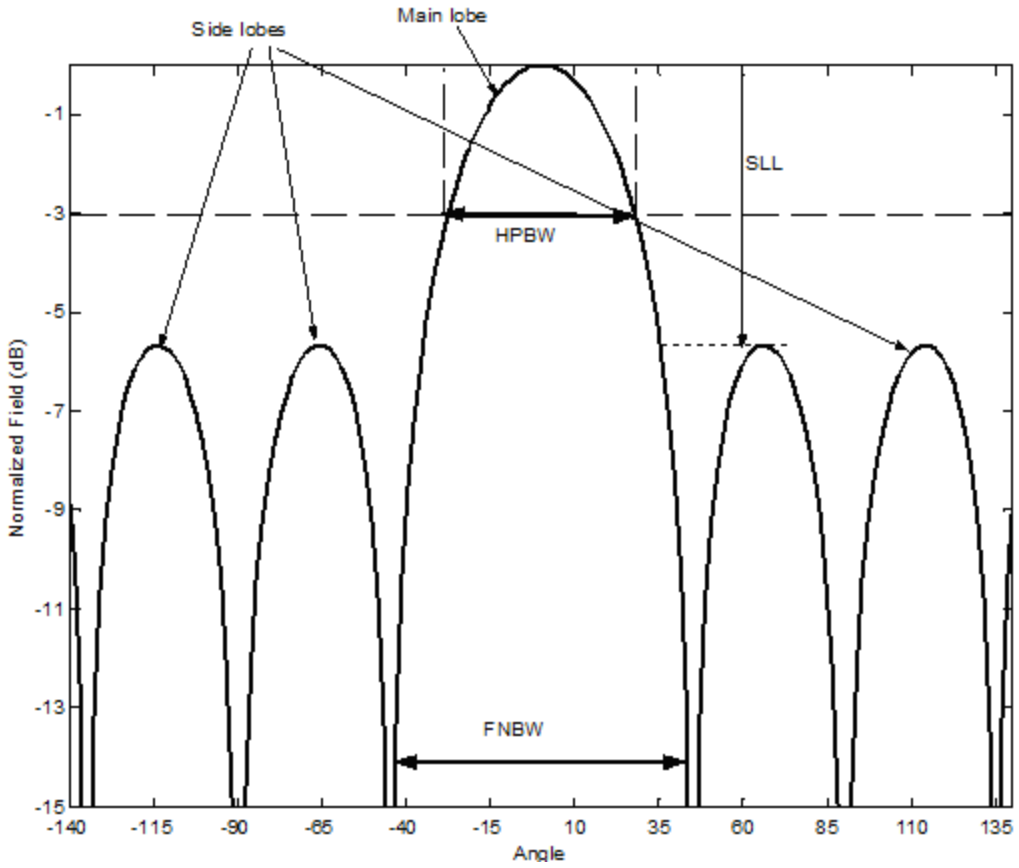
Radiation pattern



Patterns
- Power
- Electric
- Magnetic

E or H plane

Beamwidth

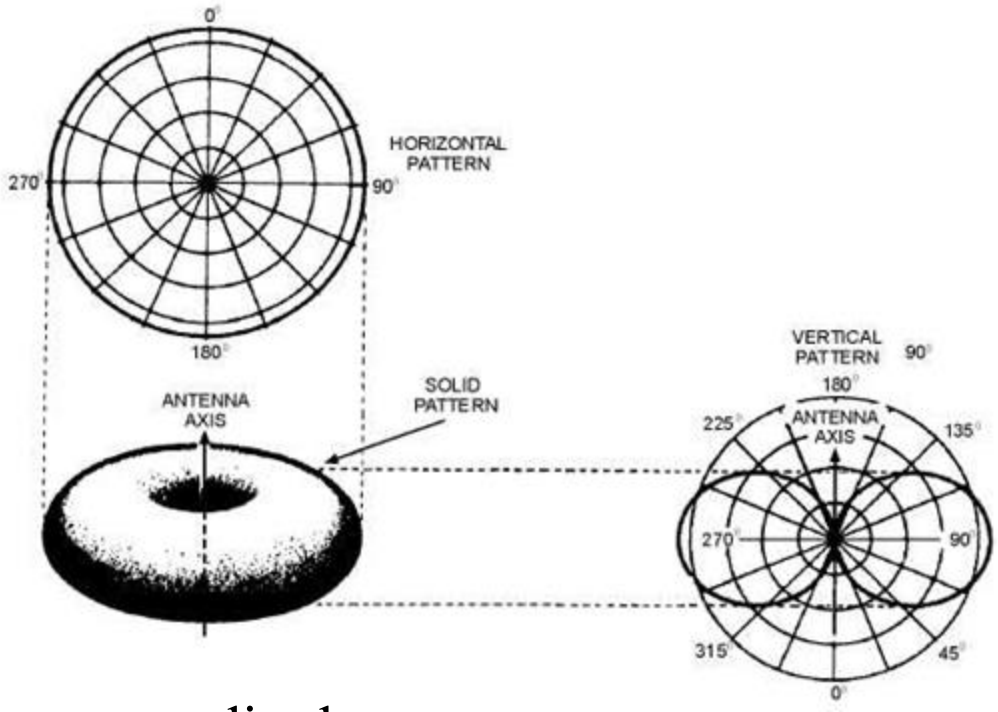


SLL: Sidelobe level

FNBW: First Null Beamwidth

HPBW: Half Power Beamwidth

3D radiation patterns [internet]



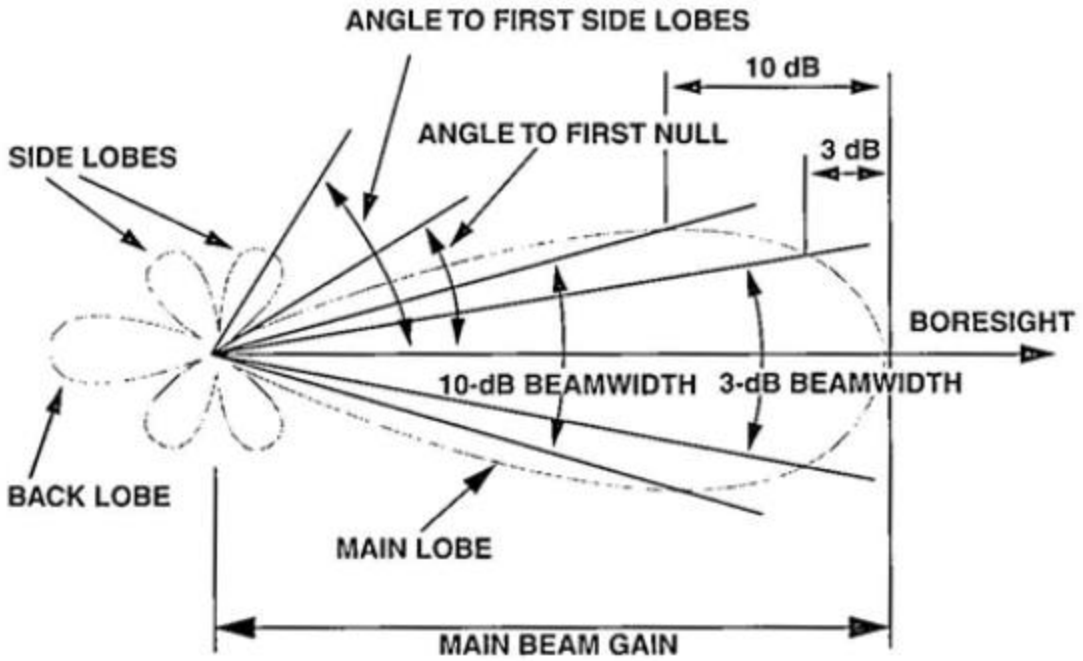
dipole



dish

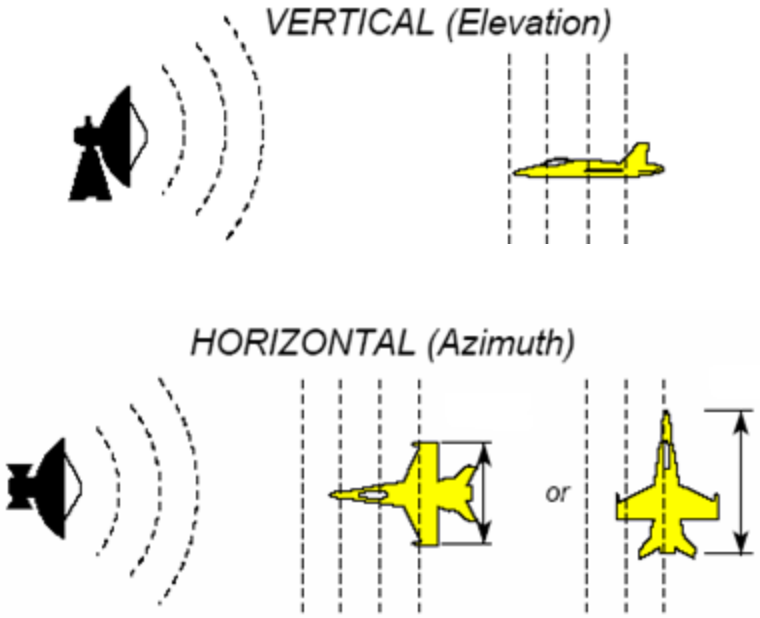
Antenna pattern: main-side lobes



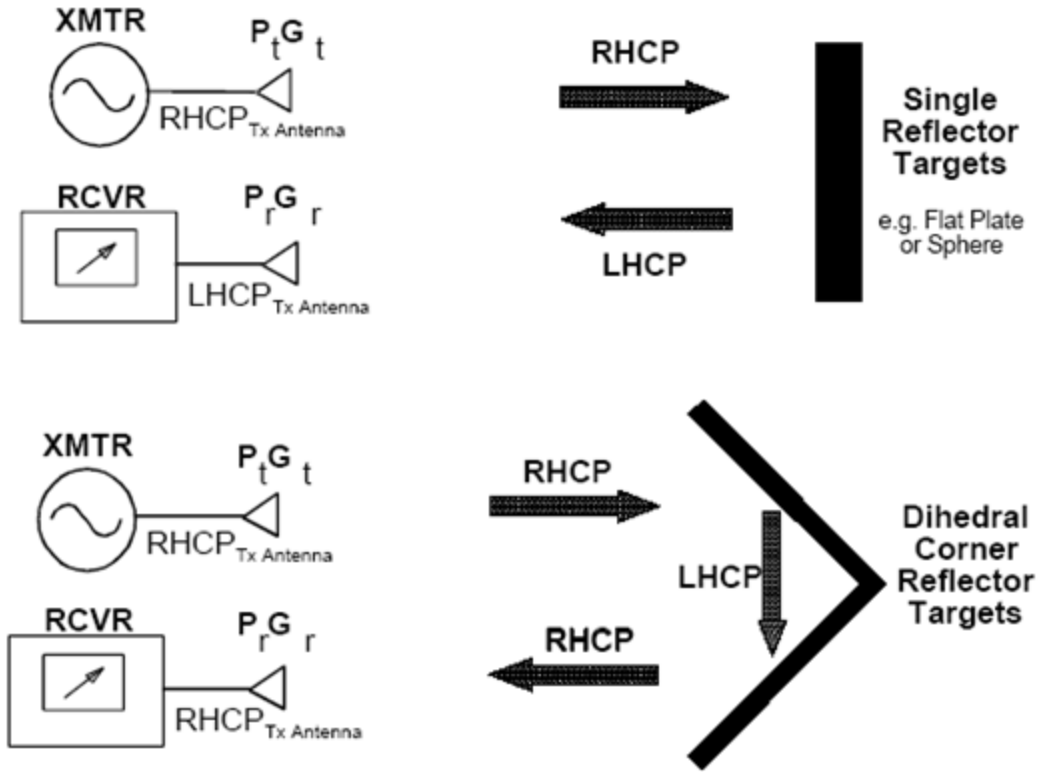


[Adamy]

Elevation and Azimuth (planes)



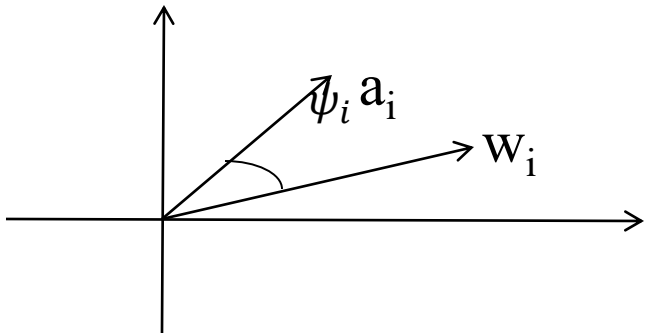
Polarization change



Polarization Loss Factor (PLF)

The angle between
- the incident wave polarization (w_i) and
- antenna polarization (a_i)

$$PLF = |\cos\psi_i|^2$$
$$PLF(dB) = 10\log PLF$$



Directivity and Gain

Directivity: the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions

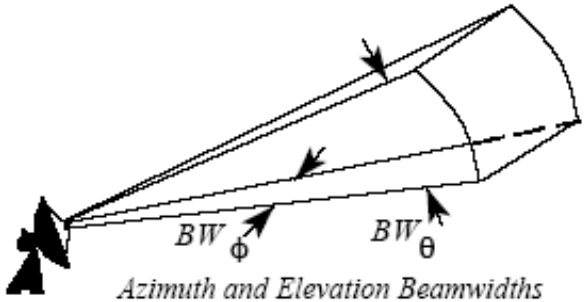
$$\text{Directivity} \equiv D = \frac{max[S(\theta,)]}{P_t / 4\pi R^2}$$

$S(\theta,)$*:Radiation intensity*

P_t*:radiated power*

Gain$=$

η*: efficiency*$(\%55-65$ *or* $0.55-0.65$ *depending on antenna*$)$



Azimuth and Elevation Beamwidths

Antenna *effecttive area*(*radar antennas: parabolic dish, horn etc.*)*:* A_e

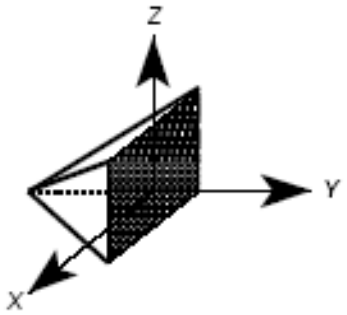
$$G = \frac{4\pi}{\lambda^2} A_e$$

G*:gain*

λ*: wavelength*

Corner Reflector and Dish antenna

CORNER REFLECTOR



Elevation: (Z-Y)
Azimuth: (X-Y)

Dependent upon feed emitter

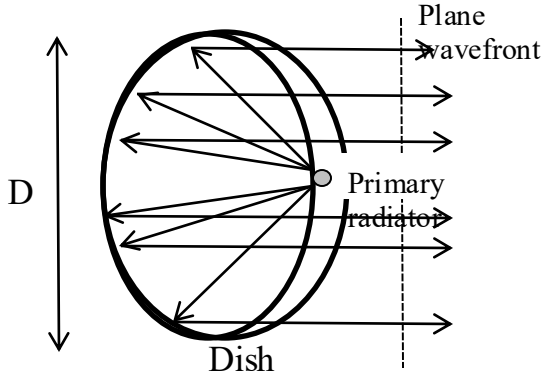
Polarization:
Feed dependent

Typical Half-Power Beamwidth
40 deg x variable

Typical Gain: *10 dB above feed*

Bandwidth: *Narrow*

Frequency Limit
Lower: *1 GHz*
Upper: *40 GHz*

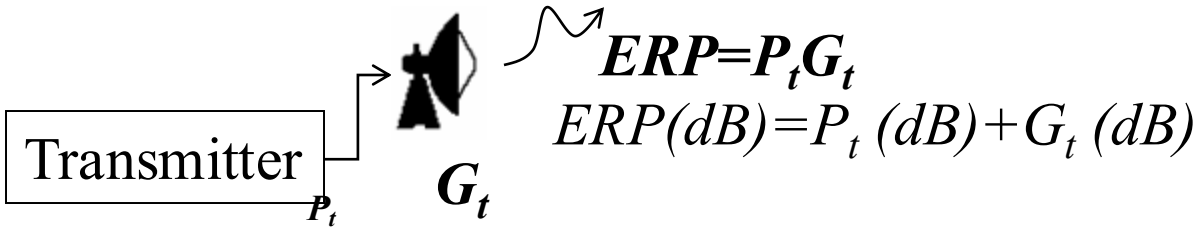


$$G=\frac{4\pi}{\lambda^2}A_e = 0.55\left(\frac{D}{\lambda}\right)^2$$ dBi,dB or dBd

$$HPBW=70\frac{\lambda}{D}\ degree$$

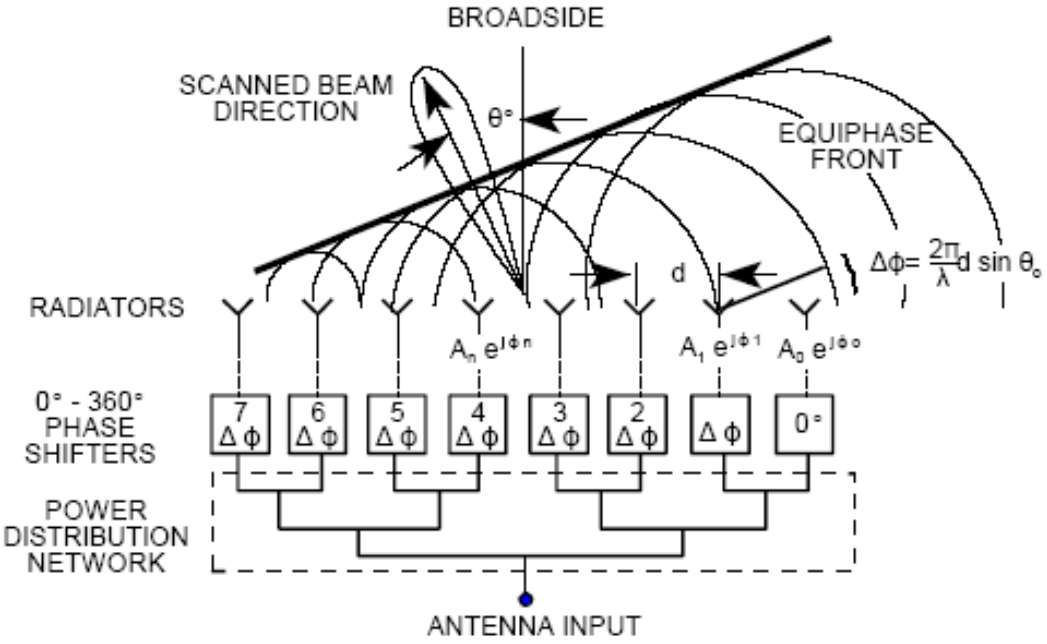
Effective Radiated Power (ERP)

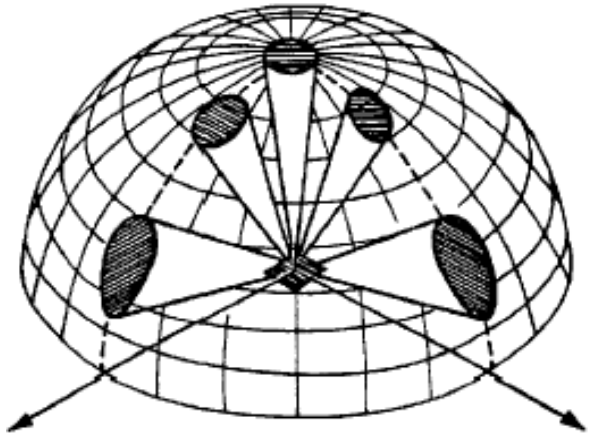
Reference in gain is half-wave dipole (1.64 or 2.15 dB)



EIRP: Effective Isotropic Radiated Power
(gain is relative to an isotropic radiator)

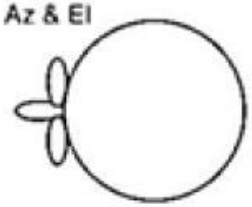
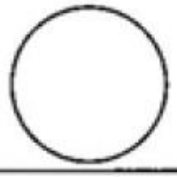
Arrays

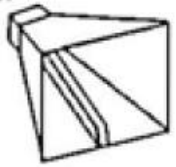
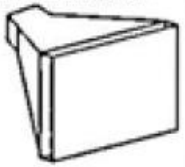
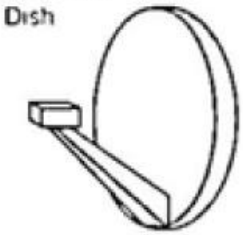
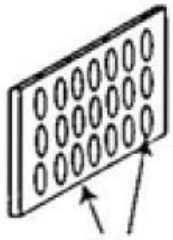
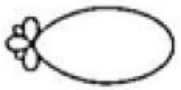




Beam scanning (distortion)

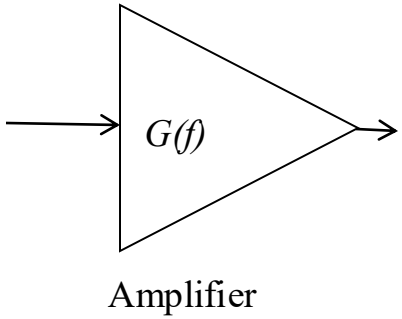
Antenna (one shot) [Adamy]

Antenna Type	Pattern	Typical Specifications
Cavity Backed Spiral 	Az & El 	Polarization: R & L Horizontal Beamwidth 60° x 60° Gain: -15 dB (min freq) +3 dB (max freq) Bandwidth 9 to 1 Frequency Range µw
Conical Spiral 	Az & El 	Polarization: Circular Beamwidth: 60° x 60° Gain: 5 to 8 dB Bandwidth: 4 to 1 Frequency Range: UHF through µw
4 Arm Conical Spiral 	El  Az 	Polarization: Circular Beamwidth: 50° x 360° Gain: 0 dB Bandwidth: 4 to 1 Frequency Range: UHF through µw

Horn 	El  Az 	Polarization: Linear Beamwidth 40° x 40° Gain 5 to 10 dB Bandwidth 4 to 1 Frequency Range: VHF through mmw
Horn with Polarizer 	El  Az 	Polarization: Circular Beamwidth 40° x 40° Gain: 4 to 10 dB Bandwidth: 3 to 1 Frequency Range: µw
Parabolic Dish 	Az & El 	Polarization: Depends on Feed Beamwidth: 0.5° x 30° Gain: 10 to 55 dB Bandwidth: Depends on Feed Frequency Range: UHF to µw
Phased Array  Elements	El  Az 	Polarization: Depends on Elements Beamwidth: 0.5° x 30° Gain: 10 to 40 dB Bandwidth Depends on Elements Frequency Range: VHF to µw

Noise Figure (NF): detection systems

Every device injects some noise into the system
NF: metric showing how noisy the device is



Amplifier

$$NF = 10log\left(\frac{S_{in}/N_{in}}{S_{out}/N_{out}}\right)$$

S_i/N_i *: input signal to noise power ratio*

S_o/N_o *: output signal to noise power ratio*

$$Ideal\ device\ (noiseless): NF = 1 = 0\ dB$$

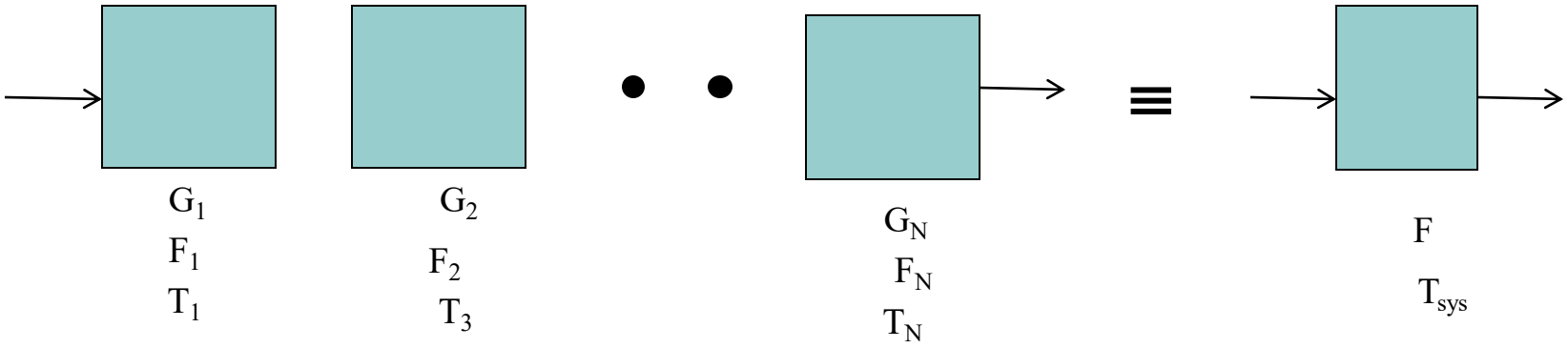
Equivalent Noise Temperature ⇔Noise Figure

- Low-noise systems such as front end of radar receivers, LNAs in a satellite receiver, noise figure is often very close to 0 dB (NF=1) (several dB -> 3-5 dB)
- Preferable to use equivalent noise temperature instead of noise figure

$$NF = \frac{G(kT_e + T_0)}{GkT_0} = \frac{T_e + T_0}{T_0}$$

$$Equivalent\ noise\ temperature$$
$$T_e = (NF - 1)T_0$$

Noise in Cascaded systems



$$NF = F_1 + \frac{NF_2 - 1}{G_1} + \frac{NF_3 - 1}{G_1 G_2} + \ldots$$

$$T_{sys} = {}_1 + \frac{T_2 - 1}{G_1} + \frac{T_3 - 1}{G_1 G_2} + ..$$